FILIP KOVAČEVIĆ

PhD Candidate in Machine Learning, ISTA
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RESEARCH INTERESTS

Learning theory, high-dimensional statistics; sample efficient learning - data selection, transfer learning, optimization.

WORK AND RESEARCH EXPERIENCE

Flatiron Institute, Visiting Researcher — New York, Jul 2026 - Aug 2026
Research visit focused on hyperparameter transfer; delivered an invited lecture on heterogeneous sample selection.

Department of Computer Science, ETH Zürich, Research Assistant — Zürich, Feb 2024 - Aug 2024
Studied high-dimensional fairness–accuracy trade-off and regularization effect on Pareto frontier; published at AISTATS.

Department of Mathematics, ETH Zürich, Teaching Assistant — Zürich, Feb 2022 - Jan 2024
Prepared and held exercise sessions for the courses Linear Algebra, Group Theory, and Galois Theory.

Microsoft, Software Engineer Intern — Belgrade, Jul 2020 - Oct 2020
Prototyped integration of an online ML audio-denoising component into a video player; contributed to problem preparation and review for the Bubble Cup programming competition.

EDUCATION

Institute of Science and Technology Austria — 2024 - present
PhD in Computer Science,
Supervisor: Prof. Marco Mondelli

Department of Mathematics, ETH Zürich — 2021 - 2024
Master's in Mathematics with Distinction | GPA: 5.81/6.00
- *ETH-D, merit-based scholarship covering full study and living costs*

Faculty of Mathematics, University of Belgrade — 2017 - 2021
Bachelor of Mathematics | GPA: 10.0/10.0
- *Award for Best Students of Faculty of Mathematics*
- *Scholarship for Exceptionally Gifted Students, national merit-based scholarship*

PUBLICATIONS AND PREPRINTS

Preprint	High-Dimensional Asymptotics and Dataset Selection for Private Transfer Learning ***F. Kovačević**, E. Cyffers, S. S. Mannelli, M. Mondelli*	2026
ICML	Full-Batch Gradient Descent Outperforms One-Pass SGD: Sample Complexity Separation in Single Index Learning ***F. Kovačević**, H. C. Ji, D. Wu, M. Soltanolkotabi, M. Mondelli*	2026
ICML	Towards a Holistic Understanding of Selection Bias for Causal Effect Identification *Y. Qiu, **F. Kovačević**, S. Huang, P. Spirtes, F. Locatello*	2026
ICLR, Oral	High-dimensional Analysis of Synthetic Data Selection *P. Rezaei, **F. Kovačević**, F. Locatello, M. Mondelli*	2026
COLT	Spectral Estimators for Multi-Index Models: Precise Asymptotics and Optimal Weak Recovery ***F. Kovačević**, Y. Zhang, M. Mondelli*	2025
AISTATS	Learning Pareto Manifolds in High Dimensions: How Can Regularization Help? *T. Wegel, **F. Kovačević**, A. Tifrea, F. Yang*	2025

SELECTED TALKS

Flatiron Institute	Selective Use of Heterogeneous Data	New York, 2026
ICLR	High-dimensional Analysis of Synthetic Data Selection	Rio de Janeiro, 2026
Youth in High-Dimensions	Spectral Estimators for Multi-Index Models	Trieste, 2025

SKILLS

Technical: Python (NumPy, PyTorch, scikit-learn), C.
Languages: Serbian (native); English (fluent, C2).